Chenru Wen

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Research Interests

I work on robot learning for manipulation, focusing on recovery from execution failures by forecasting the outcomes of candidate actions. I am currently exploring how physical estimates and their uncertainty should enter dexterous policies.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) Aug 2025 – present
M.Sc. in Robotics. Advisors: Xingxing Zuo, Dezhen Song Abu Dhabi, UAE

- **Research Intern** at OpenDriveLab, Shanghai Innovation Institute, Jun 2026 – Sep 2026

Tsinghua University, Yao Class (IIIS) Sep 2021 – Jun 2025
B.Eng. in Computer Science and Technology Beijing, China

- **Visiting Research Intern** at UC San Diego, Feb 2024 – Aug 2024

Publications

ForkCast: Predictive Recovery for Vision-Language-Action Policies via Counterfactual Futures

Chenru Wen*, Huajian Zeng*, Haokai Ding, Jian Kang, Anqing Duan, Dawei Zhou, Dezhen Song, Xingxing Zuo

Under review at IEEE International Conference on Robotics and Automation (ICRA), 2027.

arxiv.org/abs/2610.18650

Lay-Your-Scene: Natural Scene Layout Generation with Diffusion Transformers

Divyansh Srivastava, Xiang Zhang, He Wen, **Chenru Wen**, Zhuowen Tu

In IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

arxiv.org/abs/2505.04718

Research Experience

Predictive recovery for frozen VLA policies (ForkCast) Mar 2026 – Sep 2026
RCL, MBZUAI. Advisor: Xingxing Zuo ICRA 2027, under review

- **Research Topics:** vision-language-action policies, failure recovery, latent world models, active perception.
- Proposed ForkCast, a recovery framework that decides whether and how to intervene in a frozen VLA policy by comparing forecast futures of continuing the policy, re-observing with a moving camera, and replanning the arm.
- Designed a candidate-conditioned latent world model with a shared progress head that scores heterogeneous candidates at a common horizon; trained on branched rollouts, it transfers to two other VLA backbones without retraining.
- Built the real-robot pipeline on a dual-arm platform. ForkCast improved a frozen $\pi_{0.5}$ under injected occlusion and slip in simulation, and in both clean and visibility-limited settings on the real robot.

Localization on non-metric maps Oct 2025 – present
MBZUAI. Advisor: Dezhen Song In preparation

- **Research Topics:** visual localization, non-metric maps, spatial reasoning.
- Proposed localizing a walking camera on a printed visitor map, without GPS or a metric map, by enumerating map alignments consistent with local odometry and the bearings of a few named landmarks.
- Analyzed which map-side constraints carry information for localization, and found that most are necessary but uninformative, and that ambiguity in the reference itself bounds what accuracy can be measured.

Causal analysis of appearance shifts in VLA policies Jun 2026 – Sep 2026
Research Intern, OpenDriveLab, Shanghai Innovation Institute Shanghai, China

- **Research Topics:** VLA robustness, mechanistic interpretability, causal intervention.
- Turned a real-robot failure, where changing only a target's appearance changed the grasp, into controlled matched-state experiments with a fine-tuned $\pi_{0.5}$.
- Traced the effect through the visual encoder, VLM and action expert with probing and activation replacement, and showed that replacing full visual representations restores the original actions while finer interventions do not.

Engineering

Real-robot VLA platform (dual-arm AgileX PiPER) Dec 2025 – Sep 2026
RCL, MBZUAI. Used for π_0 / $\pi_{0.5}$ deployment and all ForkCast real-robot experiments Abu Dhabi, UAE

- **Hardware.** Dual PiPER arms; a RealSense head camera on an IQR PTU2 pan/tilt unit mounted on a lift that also carries both arm bases; wrist cameras; a U-Arm leader arm for teleoperation.
- **System engineering.** Adapted openpi to the robot: a ROS 2 inference client, a Docker policy server, and real-time action chunking. Wrote the pan/tilt and lift control, a library of camera views, and the ForkCast selector service.
- **Data and evaluation.** Collected demonstrations by teleoperation and replay with one-command collection and inference scripts, and ran every real-robot trial reported in ForkCast.

Honors and Awards

Lenovo Scholarship, Tsinghua University 2024
First-Class Freshman Scholarship, Tsinghua University 2021

Service

Reviewer: IEEE International Conference on Robotics and Automation (ICRA) 2027.

Skills

Programming	Python, C/C++, Bash
Learning	PyTorch; RL for dexterous hands; VLA fine-tuning (π_0, $\pi_{0.5}$); latent world models
Simulation	Isaac Lab, ManipTrans / OakInk2, RoboTwin 2.0 (SAPIEN)
Robots	AgileX PiPER dual-arm, pan/tilt and lift camera rigs, Intel RealSense